STAT UNIT 3 - PLAIN-ENGLISH EXAM CHEAT SHEET

Know what the formula is doing - not just what buttons to press
Ch. 12 Probability + Ch. 13 Proportion Inference

READ THE WORDS -> PICK THE MODEL -> DEFINE THE LETTERS -> CHECK CONDITIONS -> CALCULATE -> EXPLAIN IN CONTEXT

START HERE: SYMBOL DECODER

P(A) = chance event A happens. **A or B** = either one or both. **A and B** = both happen. **P(B|A)** = chance of B among cases where A already happened ('B given A').
n = number tried; **x** = number of successes; **p** = true probability/proportion; **p-hat=x/n** = proportion found in your sample; **q=1-p**.
μ = average; **σ** = standard deviation (typical distance from average); **E** = margin of error; **C** = confidence level; **z*** = confidence multiplier.

WHICH TOOL DO I USE?

Count arrangements? counting. **One/two events?** probability rules. **Repeated yes/no trials?** binomial. **Arrivals per time/space?** Poisson. **Chance for a sample proportion?** sampling distribution. **Estimate true proportion?** confidence interval.

1. BASIC PROBABILITY

WHAT IT MEANS: Probability is a number from 0 (impossible) to 1 (certain). The sample space is every possible outcome.
DO THIS: For equally likely outcomes: **P(event)=# wanted outcomes / # total outcomes**. For observed data: **# times it happened / # trials**.

NOT / complement: P(not A)=1-P(A).
At least one: 1-P(none).

OR, AND, AND GIVEN

USE WHEN: The question says **OR**.
WHAT IT MEANS: Add the two chances, then subtract the overlap because it was counted twice.

P(A or B)=P(A)+P(B)-P(A and B)

If A and B cannot happen together (disjoint), overlap=0.
USE WHEN: The question says **AND**.
WHAT IT MEANS: Multiply the first chance by the second chance after the first event.

P(A and B)=P(A) × P(B|A)

If events are independent, the first does not change the second, so multiply P(A)×P(B). Without replacement usually changes the second chance.
USE WHEN: The question says **given that**.

P(B|A)=P(A and B)/P(A)
Read it as: 'both A and B' divided by 'all A.'

TRAP: P(B|A) and P(A|B) are different. The event after the | is the group you restrict to.

BAYES IN PLAIN ENGLISH

USE WHEN: You know test accuracy or conditional rates and must work backward, such as 'positive test -> actually sick?'
DO THIS: Make two paths: desired group and other group. Multiply along each path. Then divide the desired positive path by **all** positive paths.

P(B|A)= desired A-and-B path / all paths ending in A

Example: flu=.10; P(+|flu)=.95; P(+|no flu)=.01.
Desired=.095. All positive=.095+.009=.104.
Answer=.095/.104=.913.

2. COUNTING

DO THIS: For several choices in a row, multiply the choices at each step.
USE WHEN: Permutation: order or jobs matter (lineup, officers, ranked places).
nPr=n!/(n-r)!
USE WHEN: Combination: only the group matters (team, committee, choose flavors).
nCr=n!/[r!(n-r)!]

Repeated identical objects: $n!/(r_1!r_2!\ldots)$.
Probability from counting = wanted arrangements / total arrangements.

3. RANDOM VARIABLES

WHAT IT MEANS: A discrete random variable is a count such as # defects. A probability table must have probabilities between 0 and 1 that add to 1.
DO THIS: To find the long-run average, multiply every possible value by its probability, then add.

Expected value: μ=Σ[x·P(x)]

DO THIS: Variance measures spread. Square each value first, weight by probability, add, then subtract the mean squared.

Variance $\sigma^2=\Sigma[x^2P(x)]-\mu^2$; SD $\sigma=\sqrt{\sigma^2}$

4. BINOMIAL: IS IT THE RIGHT MODEL?

USE WHEN: You count successes in a fixed number of repeated yes/no trials.
Check **BINS**: **B**inary outcomes; **I**ndependent trials; fixed **N**umber of trials; **S**ame success chance p each time.
n=# trials; x=# successes wanted; p=success chance; q=1-p=failure chance.
WHAT IT MEANS: nCx counts the different placements of the x successes. p^xq^{n-x} is the chance of one such placement.

$P(X=x)=nCx\cdot p^x\cdot q^{n-x}$
mean=np; variance=npq; SD=√(npq)

Example: exactly 3 heads in 5 flips -> n=5, x=3, p=.5, q=.5.
TRAP: Not binomial if you keep going until success, p changes, or trials depend on earlier results.

BINOMIAL CALCULATOR TRANSLATOR

binompdf(n,p,x) = exactly x.
binomcdf(n,p,x) = x or fewer (≤x).

at most b -> cdf(b)
less than b -> cdf(b-1)
at least a -> 1-cdf(a-1)
more than a -> 1-cdf(a)
a through b -> cdf(b)-cdf(a-1)
TI-84: **2nd -> VARS (DISTR)**. Equality decides whether you shift the cutoff by 1. Example above: binompdf(5,.5,3).

5. POISSON

USE WHEN: Count arrivals/events in a fixed time or space when events are independent and the average rate stays constant.
WHAT IT MEANS: λ (lambda) is the expected count for the interval in the question. Scale it first: rate × interval length.

$P(X=x)=e^{-\lambda}\lambda^x/x!$
mean=λ; variance=λ; SD=√λ

Example: 2 customers per 15 min for 1 hour -> λ=2×4=8.
TI-84: poissonpdf(λ,x)=exactly x; poissoncdf(λ,x)=x or fewer. Use the same word/cutoff rules as binomial.
Poisson shortcut for binomial (worksheet): n≥100 and np≤10; set λ=np.

6. BELL-CURVE SHORTCUT FOR BINOMIAL

USE WHEN: n is large enough that a binomial histogram looks approximately normal. Worksheet check: np≥5 and nq≥5.
DO THIS: Use mean np and SD √(npq), but shift whole-number boundaries by 0.5 so the continuous bell curve covers the correct bars.

exactly c: c-0.5 to c+0.5
at most c: upper c+0.5
less than c: upper c-0.5
at least c: lower c-0.5
more than c: lower c+0.5
a through b: a-0.5 to b+0.5

Then use normalcdf(lower,upper,mean,SD).

7. WHY SAMPLE PROPORTIONS VARY

WHAT IT MEANS: Different random samples give slightly different p-hat values. The sampling distribution describes that repeated-sampling wiggle.

Center: $\mu_{p\text{-hat}}$=p
Typical wiggle (SE): √[p(1-p)/n]
z=(observed p-hat - p)/SE

Example: if p=.14 and n=128, center=.14 and SE=√(.14×.86/128)≈.031. A sample p-hat commonly misses .14 by roughly .031.
DO THIS: For a probability about p-hat, use the true p in the SE. Convert the cutoff to a z-score, then find left/right/between area.
Conditions from slides: random sample/assignment; n<10% of population; expected successes np≥10 and failures n(1-p)≥10.
WHAT IT MEANS: Bigger n means less wiggle and a more precise sample, but the center stays p.

Course note: Pearson uses 5/5 for its normal-approximation exercise; instructor slides use 10/10 for proportion inference.

8. CONFIDENCE INTERVAL FOR A PROPORTION

USE WHEN: You have one sample and want a believable range for the true population proportion p.
WHAT IT MEANS: Start at what your sample found (p-hat). Move left and right by the margin of error E.

p-hat=x/n
SE=√[p-hat(1-p-hat)/n]
E=z*×SE
CI=(p-hat-E, p-hat+E)

z* tells how many SEs to go out: 90%=1.645; 95%=1.960; 98%=2.326; 99%=2.576.
Conditions (slides): randomization; n<10% of population; observed successes x≥10 and failures n-x≥10.
TI-84: **STAT -> TESTS -> 1-PropZInt**; enter x, n, C-Level.
Course example: 93/131=.710. The 95% CI (.632,.788) means the believable range for the true town proportion is 63.2% to 78.8%.

HOW TO WRITE THE CONCLUSION

Correct: 'We are 95% confident that the interval from L to U captures the true proportion of [population] who [response].'
WHAT IT MEANS: If we repeatedly sampled the same way, about 95% of the intervals would capture the fixed true p.
TRAP: After the interval is calculated, do not say there is a 95% probability p is inside it. Do not describe other samples.

MINIMUM SAMPLE SIZE

USE WHEN: The question gives desired margin of error E but not n.

n=(z*/E)²·p*(1-p*)

Use an earlier estimate for p*. If none is given, use .5. **Always round n up.** Higher confidence or smaller E needs more people. Halving E needs about 4 times n.

9. CONFIDENCE INTERVAL FOR VARIANCE/SD

USE WHEN: Estimate population variance σ² or standard deviation σ. Population must be approximately normal.
df=n-1; α=1-C. From a right-tail chi-square table, χ^2_R uses area α/2 and is the larger cutoff; χ^2_L uses area 1-α/2 and is smaller.

variance lower=$(n-1)s^2/\chi^2_R$
variance upper=$(n-1)s^2/\chi^2_L$

For an SD interval, square-root both endpoints. Chi-square is lopsided, so do **not** write estimate ± margin of error.

FINAL POINT-SAVERS

1) Circle words: or, and, given, at least, more than, without replacement.
2) Name the model and define n, x, p/λ before typing.
3) Check conditions before inference.
4) Keep full calculator precision; round only the final answer.
5) Give conclusions in the problem's real-world context.